A Second-Order Perspective on Model Compositionality and Incremental Learning

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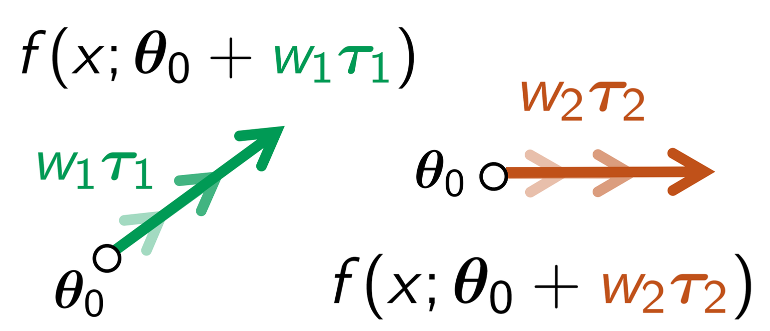
ICLR

Multi-tasking by Model Compositionality

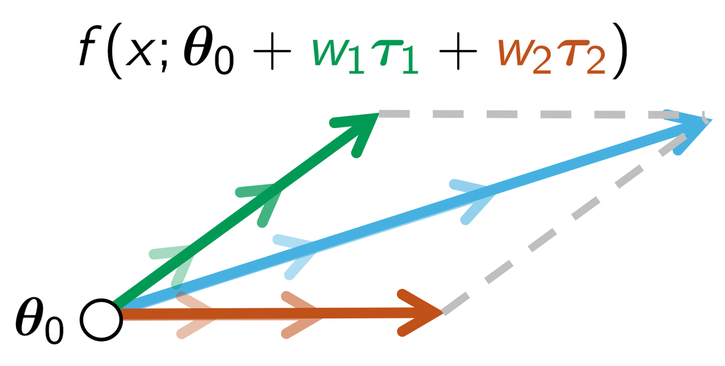
We consider a pool $\mathcal{P} = \{f(\cdot;\boldsymbol{\theta}_t) \mid \boldsymbol{\theta}_t \triangleq \boldsymbol{\theta}_0 + \boldsymbol{\tau}_t\}_{t=1}^{T}$ of models fine-tuned from a common set of pre-training weights $\boldsymbol{\theta}_0$.

- Each model is fine-tuned **on a separate, distinct task** $\mathcal{D}_t$.



The *vector* $\tau_t = \theta_t - \theta_0$ (**task vector**) indicates the displacement w.r.t. θ_0 (**parameter space**) after training on task t.

We obtain the weights of the **composed model** $f_{\mathcal{P}}$ by averaging the weights within the pool.



$$f_{\mathcal{P}} \triangleq f(\cdot;\boldsymbol{\theta}_{\mathcal{P}}) \quad \sum_{t=1}^{T} w_t = 1$$

$$\boldsymbol{\theta}_{\mathcal{P}} = \boldsymbol{\theta}_0 + \sum_{t=1}^{T} w_t\boldsymbol{\tau}_t$$

$$\hat{\ell}(\boldsymbol{\theta}|\mathcal{D}) = \frac{1}{\sum_{t=1}^{T} n_t} \sum_{\boldsymbol{x},\boldsymbol{y}\in\mathcal{D}} \ell(\boldsymbol{\theta}|\boldsymbol{x},\boldsymbol{y})$$

Research questions

Given the risk $\hat{\ell}(\boldsymbol{\theta}_t|\mathcal{D})$ of each individual model $f(\cdot;\boldsymbol{\theta}_t)$, what can we say about that of the merged model $f(\cdot;\boldsymbol{\theta}_{\mathcal{P}})$?

Individual training. How can we train individual learners $f(\cdot;\boldsymbol{\theta}_t)$ on distinct tasks to still achieve a composition $f(\cdot;\boldsymbol{\theta}_{\mathcal{P}})$ with low global risk?

Ensemble training. Instead of optimizing each model on its individual loss, could we optimize each model based on the loss of the whole composed model $f(\cdot;\boldsymbol{\theta}_{\mathcal{P}})$?

We use the **second-order** Taylor approximation of the risk around θ_0:

$$\ell_{\text{cur}}(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}_0) + (\boldsymbol{\theta}-\boldsymbol{\theta}_0)^{\mathsf{T}}\nabla\ell(\boldsymbol{\theta}_0) + \frac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)^{\mathsf{T}}\mathbf{H}_{\ell}(\boldsymbol{\theta}_0)(\boldsymbol{\theta}-\boldsymbol{\theta}_0).$$

Jensen's inequality. Assuming that the Hessian $\mathbf{H}_{\hat{\ell}}(\boldsymbol{\theta}_0) \succeq 0$ is positive semidefinite (*e.g.*, when $\boldsymbol{\theta}_0$ is a local minimum of $\hat{\ell}(\boldsymbol{\theta})$), we can say:

$$\hat{\ell}_{\text{cur}}(\boldsymbol{\theta}_{\mathcal{P}} = \boldsymbol{\theta}_0 + \textstyle\sum_{t=1}^{T} w_t\boldsymbol{\tau}_t) \leq \sum_{t=1}^{T} w_t\, \hat{\ell}_{\text{cur}}(\boldsymbol{\theta}_t = \boldsymbol{\theta}_0 + \boldsymbol{\tau}_t)$$

If each model is trained to the global optimum ($\rightarrow$ low risk across all tasks), then there are guarantees on the risk of the composed model.

Individual training. The t^{th} model $f(\cdot;\boldsymbol{\theta}_t)$ trains on $\mathcal{D}_t \sim p_t(\boldsymbol{x},\boldsymbol{y})$ only.

We expect:
- For $x, y \in p_t(\boldsymbol{x},\boldsymbol{y})$: *low* risk.
- For $x, y \in p_{t'\neq t}(\boldsymbol{x},\boldsymbol{y})$: ***high*** risk.

$\rightarrow$ *low* OOD performance.

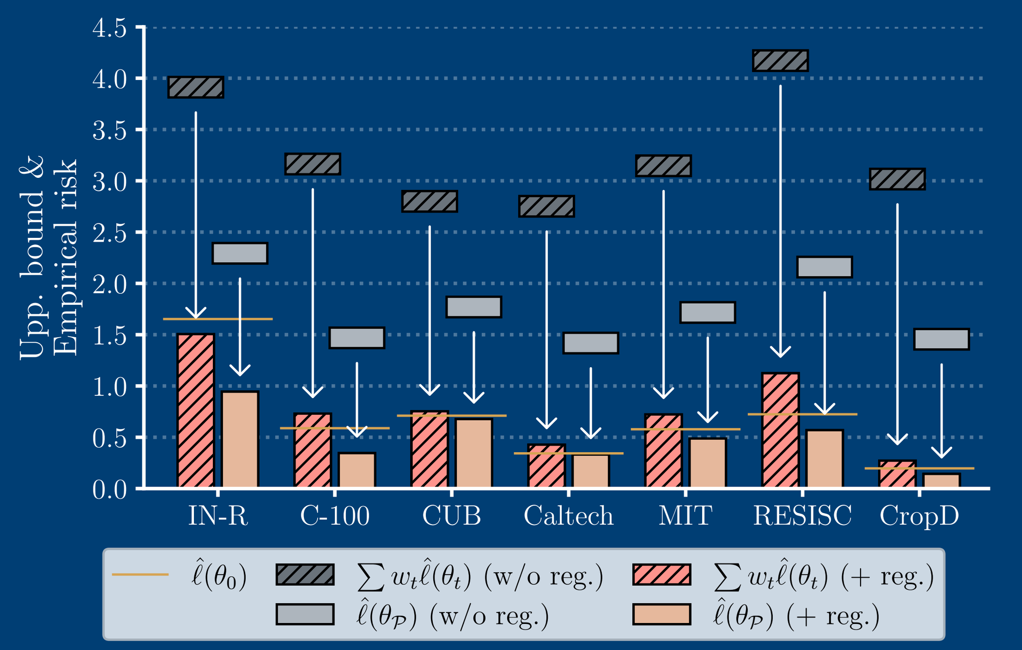
⚠ The right-hand side of Jensen's inequality grows large, losing its effectiveness as an upper bound for the composed model.

Idea. Reduce the upper bound through **explicit regularization.**

For external data, we minimize:

$$\mathcal{D}_{\text{KL}}(p_{\boldsymbol{\theta}_0}(\boldsymbol{y}|\boldsymbol{x})||p_{\boldsymbol{\theta}_t}(\boldsymbol{y}|\boldsymbol{x}))$$

using the pre-training output distribution as anchor for the t^{th} model.



Incremental Task Arithmetic (ITA)

$$\min_{\boldsymbol{\theta}_t} \ \mathbb{E}_{p_t(\boldsymbol{x},\boldsymbol{y})}[\ell_{\text{cur}}(\boldsymbol{\theta}_t|\boldsymbol{x},\boldsymbol{y})] + \text{EWC}_{\boldsymbol{\theta}_0}$$

$$\text{EWC}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}_t) = \sum_i^{|\theta|} \hat{\text{F}}_{\boldsymbol{\theta}_0}^{(i)}(\theta_t^{(i)} - \theta_0^{(i)})^2.$$

Ensemble training. Train each model to minimize the risk of the composed model $f(\cdot;\boldsymbol{\theta}_{\mathcal{P}})$ directly (**ensemble training**). Doing the math:

$$\ell_{\text{cur}}(\boldsymbol{\theta}_{\mathcal{P}}) + \Omega(\boldsymbol{\theta}_1,\ldots,\boldsymbol{\theta}_T) = \sum_{t=1}^{T} w_t \ell_{\text{cur}}(\boldsymbol{\theta}_t)$$

$$\textbf{where} \quad \underbrace{\Omega(\boldsymbol{\theta}_1,\ldots,\boldsymbol{\theta}_T)}_{\text{Alignment term}} = \frac{1}{2}\sum_{t=1}^{T}\sum_{t'<t} w_t w_{t'}(\boldsymbol{\tau}_t - \boldsymbol{\tau}_{t'})^{\mathsf{T}}\mathbf{H}_{\ell}(\boldsymbol{\theta}_0)(\boldsymbol{\tau}_t - \boldsymbol{\tau}_{t'}).$$

Incremental Ensemble Learning (IEL)

$$\underset{\boldsymbol{\theta}_t}{\text{minimize}} \quad \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim p_t(\boldsymbol{x},\boldsymbol{y})}[\ell_{\text{cur}}(\boldsymbol{\theta}_{\mathcal{P}}|\boldsymbol{x},\boldsymbol{y})] + \beta\ \Omega_{\hat{\text{F}}}(\boldsymbol{\theta}_1,\ldots,\boldsymbol{\theta}_t).$$

Mammoth
A PyTorch benchmark library for **Continual Learning** with **20+** datasets and **50+** methods.

for each task $t \in \{1, 2, \ldots, T\}$ **do**
  $\boldsymbol{h}_0^{(t)} \leftarrow$ Linear Probing on D_t with $f(\cdot;\boldsymbol{\theta}_0)$
  $\boldsymbol{\theta}_0 \leftarrow \boldsymbol{\theta}_0 \cup \boldsymbol{h}_0^{(t)}$ ▷ add the pre-training head
  $\hat{\text{F}}_{\boldsymbol{\theta}_0} \leftarrow \hat{\text{F}}_{\boldsymbol{\theta}_0} + \hat{\text{F}}_{\boldsymbol{\theta}_0}^{(t)}$ ▷ update the global Fisher
  Fine-tuning
  $\boldsymbol{\tau}_t \leftarrow \mathcal{N}(\mu_{\text{init}},\ \sigma_{\text{init}})$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{f(\cdot;\boldsymbol{\theta}_t = \boldsymbol{\theta}_0 + \boldsymbol{\tau}_t)\}$ ▷ extend $\mathcal{P}$
 for each example $(\boldsymbol{x},\boldsymbol{y})$ **in** D_t **do**
   $p_{\boldsymbol{\theta}}(\boldsymbol{y}|\boldsymbol{x}) \leftarrow f(\boldsymbol{x};\boldsymbol{\theta}_t)$ ▷ predict with $\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_0 + \boldsymbol{\tau}_t$
   $p_{\boldsymbol{\theta}}(\boldsymbol{y}|\boldsymbol{x}) \leftarrow f(\boldsymbol{x};\boldsymbol{\theta}_{\mathcal{P}})$ ▷ use $\boldsymbol{\theta}_{\mathcal{P}} \leftarrow \boldsymbol{\theta}_0 + \frac{1}{t}\sum_{t'=1}^{t}\boldsymbol{\tau}_{t'}$
   $\ell \leftarrow -\log p_{\boldsymbol{\theta}}(\boldsymbol{y}|\boldsymbol{x})$
   $\boldsymbol{\tau}_t \leftarrow \boldsymbol{\tau}_t - \mathtt{lr} \cdot \nabla_{\boldsymbol{\tau}_t}[\ell + \frac{\alpha}{2}\text{EWC}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}_t)]$ ▷ ITA reg.
   $\boldsymbol{\tau}_t \leftarrow \boldsymbol{\tau}_t - \mathtt{lr} \cdot \nabla_{\boldsymbol{\tau}_t}[\ell + \beta\Omega_{\hat{\text{F}}}(\mathcal{P})]$ ▷ IEL reg.

Results & take-home messages

Effective model averaging requires each model to perform well **on other tasks.**

Applying regularization and remaining **within the pre-training basin** is a data-free valuable strategy to improve OOD performance, and hence, model averaging.

Class-Incremental Learning. The model learns tasks of disjoint classes **sequentially**.

Model	IN-R	C-100	CUB	Caltech	MIT	RESISC	CropDis.
Joint	86.08	91.74	87.12	93.21	87.19	96.79	99.71
Finetune	22.31	21.57	29.35	44.03	18.85	12.90	20.54
EWC	58.64	73.49	40.33	73.23	64.44	58.80	70.33
LWF-MC	50.93	72.16	21.88	79.73	67.04	68.09	78.28
DER++	60.53	83.02	76.10	86.11	68.88	67.23	**98.77**
L2P	67.17	87.32	78.95	91.22	83.17	63.47	75.18
CODA	74.12	86.48	78.54	90.57	77.73	69.50	74.65
SEED	55.87	83.39	85.35	90.04	86.34	74.81	92.77
APT	65.32	86.19	69.51	87.71	75.83	49.99	59.37
InfLoRA	76.97	87.17	79.14	90.53	79.14	79.92	89.05
TMC	60.01	78.42	71.72	82.30	68.66	60.66	66.56
ITA-FFT	76.43	89.38	84.80	92.32	85.35	80.50	91.81
ITA-LoRA	77.79	89.96	85.55	92.65	**86.60**	82.00	95.85
ITA-(IA)[3]	77.04	**90.66**	**85.67**	92.67	84.74	**83.73**	95.41
IEL-FFT	**80.09**	89.38	84.89	92.23	82.79	81.42	95.83
IEL-LoRA	79.93	89.53	84.95	92.19	84.49	82.53	95.88
IEL-(IA)[3]	77.86	89.72	84.57	**92.70**	85.54	81.50	95.68

See the paper for tests on **specialization** and **unlearning**.